

Pallavi Shakya | PhD

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Summary

Computational biologist with end-to-end expertise across wet-lab and dry-lab genomics. Experienced in chromosome-scale de novo assembly using PacBio HiFi, Oxford Nanopore, Illumina, and Hi-C, paired with hands-on functional validation (RNAi, RT-qPCR, infection assays). Recent first-author work in *PLoS Pathogens* delivered the most contiguous plant-parasitic nematode assembly to date. Research spans plant pathology, nematology, agricultural biotechnology, and the development of bioinformatics tools and infrastructure.

Education

Plant Pathology <i>PhD, Davis, USA</i>	University of California, Davis 2021–2026
Plant Biotechnology <i>Master of Science (MSc), Wageningen, Netherlands</i>	Wageningen University 2018–2020
Biotechnology <i>Bachelor of Science (BSc), Kathmandu, Nepal</i>	SANN International College – Purbanchal University 2012–2016

Research Experience

UC Davis <i>PhD Researcher</i>	Davis, USA 2021–present
○ Genome Architecture: Delivered a chromosome-scale assembly of <i>Meloidogyne hapla</i> using PacBio HiFi and Hi-C, resolving complex repetitive regions and non-canonical telomeres. ○ Quantitative Trait Mapping: Built an F2 linkage map and mapped a single major-effect QTL governing avirulence on a resistant common bean cultivar, fine-mapping the locus to a narrow candidate interval. ○ Effector Discovery: Identified a secreted nematode effector that acts as a virulence factor on susceptible hosts and an avirulence determinant on resistant hosts, integrating infection phenotyping with gene-silencing validation. ○ Functional Validation: Validated candidate effectors and genes of interest using RNAi-mediated gene silencing and RT-qPCR; managed large-scale nematode infection assays to characterize virulence phenotypes. ○ Molecular Techniques: Executed high-molecular-weight (HMW) DNA and RNA extractions for long-read sequencing (PacBio/ONT) and designed specialized primers for complex genomic regions. ○ Pipeline Engineering: Developed and deployed reproducible Snakemake workflows for SV discovery and RNA-seq analysis, integrated with Docker for cross-platform portability. ○ Infrastructure Leadership: Established and managed a lab-wide Apollo genome browser instance, enabling collaborative real-time manual curation for 10+ researchers. ○ Data Stewardship: Organized analysis repositories under version control with reproducible documentation, enabling lab-wide reuse and a clean project handoff. ○ Comparative Genomics: Identified recombination hotspots linked to effector protein evolution in parasitic nematodes.	
Laboratory of Nematology, Wageningen University <i>Nematology Intern</i>	Wageningen, Netherlands 03/2020–09/2020
○ Curated and analyzed transcriptome datasets of Potato Cyst Nematodes. ○ Performed differential expression analysis and functional annotation.	

Skills

Omics: De novo assembly (PacBio, ONT, Hi-C), Annotation, Comparative Genomics, Transcriptomics (RNA-seq, IsoSeq), Phylogenetics

Data Science: Statistical analysis (PCA, clustering, linear/logistic regression), QTL mapping with phenotypic trait validation, exploratory analysis of high-dimensional genomic data; foundational ML coursework (Stanford ML certificate)

Bioinformatics: Snakemake, Docker, Conda, SLURM, Git/GitHub, HPC environments

Programming: R, Python, TensorFlow, Bash, Markdown, L^AT_EX

Molecular Biology: RNAi, Pathogen assays, RT-PCR, DNA and RNA extraction

Languages: English, Nepali, Hindi, Nepal Bhasa; Intermediate proficiency in Dutch

Publications

2026: Blundell, A. C., Shigekane-Kraft, E., Janakowski, S., Sobczak, M., Dai, D., **Shakya, P.**, et al. Resistance breaking in root-knot nematodes carries a fitness cost associated with defective feeding site development. *bioRxiv* 2026. doi:10.1101/2026.02.17.704924

2025: **Shakya, P.**, Maulana, M. I., Danchin, E. G., Voogt, M. L., Ruitenbeek, S. J. S. van de, Gimeno, J., Taranto, A. P., Blundell, A. C., Despot-Slade, E., Meštrović, N., Mota, A. Z., Dai, D., Williamson, V. M., Sterken, M. G., & Siddique, S. High-resolution genome assembly and linkage mapping in *Meloidogyne hapla* reveal non-canonical telomere repeats and recombination hotspots associated with effector proteins. *PLoS Pathog* 21(11). doi:10.1371/journal.ppat.1013706

2019: Thapa, C., **Shakya, P.**, Shrestha, R., Pal, S., & Manandhar, P. Isolation of polyhydroxybutyrate (PHB) producing bacteria, optimization of culture conditions for PHB production, extraction and characterization of PHB. *Nepal Journal of Biotechnology* 6(1), 62–68. doi:10.3126/njb.v6i1.22339

Honors & Awards

2026: Best Early Career Speaker Award – European Society of Nematologists Conference, Netherlands

2026: Cobb Student Travel Award – Society of Nematologists Conference, USA (**\$1,000**)

2026: Travel Bursary – European Society of Nematologists (**€900**)

2024: Mary Allen Travel Award – University of California, Davis, USA (**\$1,500**)

2023: Bayer CropScience Student Travel Award – Society of Nematologists Conference, USA (**\$600**)

2023: Jastro-Shields Research Award – University of California, Davis, USA (**\$2,500**)

2022: Jastro-Shields Research Award – University of California, Davis, USA (**\$3,000**)

2018: University Fund Wageningen (UFW) Fellowship – Wageningen University, Netherlands (**€38,600**)

2018: Anne van den Ban Fund Scholarship – Wageningen University, Netherlands (**€21,360**)

2018: HM Queen's Scholarship – Asian Institute of Technology, Thailand (**1,150,400 Baht; declined**)

2017: Second Best Poster Presentation Award – Biotechnology Society Nepal & National Academy of Science and Technology, International DNA Day 2017 (**Rs. 5,000**)

Oral & Poster Presentations

High-resolution genome assembly and linkage mapping in *Meloidogyne hapla* reveal non-canonical telomere repeats
Invited Speaker, Baltimore, Maryland, USA

High-resolution genome assembly and linkage mapping in *Meloidogyne hapla* reveal non-canonical telomere repeats
Plenary, Egmond aan Zee, Netherlands

A chromosome-scale genome assembly reveals novel chromosomal end repeats in a parasitic nematodeThe Biology Society of Nepal
Poster, Cold Spring Harbor Laboratory, New York, USA

Unraveling the virulence mechanism of *Meloidogyne hapla* through genomics and functional geneticsSociety of Nematology
Oral, Ohio State University, Ohio, USA

***Meloidogyne hapla* with improved assembly and genome annotation**International Congress of Nematology
Oral, Antibes, France 05/2022

Isolation of Polyhydroxybutyrate (PHB) Producing Bacteria, Optimization of Culture Conditions for PHB Production
Poster, Kathmandu, Nepal

Certifications

Stanford University (Online) <i>Machine Learning</i>	USA 2021
British Council <i>Cambridge English C1 Proficiency</i>	Kathmandu, Nepal 2017
Goethe Zentrum <i>German A1 Proficiency</i>	Kathmandu, Nepal 2017
Intrepid Nepal & Harvard Medical School <i>Advanced Hands-On Molecular Training, With focus on DNA extraction, real-time PCR, introductory bioinformatics</i>	

Leadership, Outreach & Mentorship

2026: Speaker & Event Lead, UC Davis Biodiversity Museum Day. Organized and hosted a public event on nematode biodiversity and plant health, delivering an accessible talk and facilitating Q&A with K–12 students, growers, and community members.

2026: Organizing Committee Member, UC Davis Biodiversity Museum Day. Coordinated public-facing exhibits on plant-parasitic and free-living nematodes, using live demonstrations and hands-on activities to communicate their roles in agriculture and ecosystems.

2022–2024: Volunteer & Nematode Representative, UC Davis Biodiversity Museum Day & Picnic Day. Designed and delivered interactive displays on nematode life cycles, host–parasite interactions, and crop damage, highlighting the importance of nematology for sustainable agriculture.

2022–2026: Mentorship & Training. Mentored two undergraduate researchers in plant–nematode interaction assays and nematode genomics, guiding experimental design and data interpretation. Trained two PhD students and postdoctoral researchers in RNA-seq and genome assembly workflows, building bioinformatics capacity in nematology-focused research groups.

2022: Teaching Assistant, Nematology (NEM100), UC Davis. Supported laboratory sessions, discussions, and grading, helping students connect nematode morphology, biology, and management principles to real-world plant diseases.

2023: Teaching Assistant, Animal Biology (ABI50), UC Davis. Assisted with laboratory sessions on nematode parasites of fishes and other taxa, reinforcing core concepts in host–parasite biology and animal health.

2017–2018: Team Leader, Raleigh International, Nepal. Spearheaded one year of cross-cultural agricultural development in rural communities. Mentored 14 volunteers (from the UK & Nepal) in community engagement and field safety. Facilitated livelihoods and sustainability training with smallholder farmers and local agro-industries, emphasizing soil health, crop diversification, and sustainable pest management. Strengthened local capacity through farmer-led workshops, linking field realities to long-term agricultural resilience.